1:51

MATLAB Drive > dsp6_1.m

```
clc;
clear;
close all;

%% Signal
fs = 1000;
f = 50;
t = 0:1/fs:0.1;
x = sin(2*pi*f*t);

%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};

figure;

for i = 1:length(L)

    % Quantization
    xq{i} = round(((x+1)*(L(i)-1)/2))*(2/(L(i)

    % Error & MSE
    e{i} = x - xq{i};
    mse(i) = mean(e{i}.^2);

    % Plot Quantized Signals
    subplot(4,1,i+1);
    stairs(t,xq{i},colors{i},'LineWidth',1.2);
    title(['Quantized Signal (' num2str(L(i))
    grid on;
end

% Original Signal
subplot(4,1,1);
plot(t,x,'LineWidth',1.5);
title('Original Signal');
grid on;

% Display MSE
fprintf('MSE for 8 Levels = %f\n',mse(1));
fprintf('MSE for 16 Levels = %f\n',mse(2));
fprintf('MSE for 32 Levels = %f\n',mse(3));

%% Error Plots
figure;
for i=1:3
    subplot(3,1,i);
```

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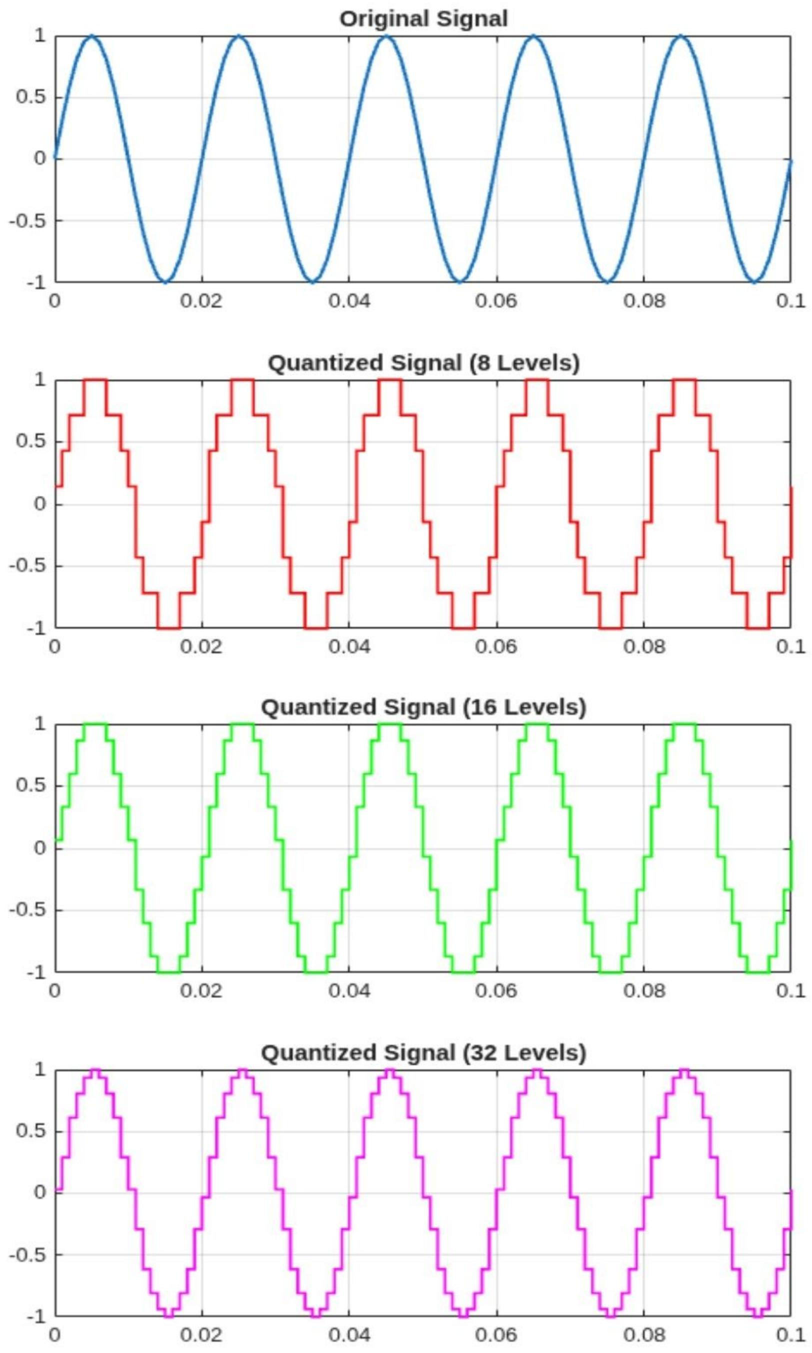
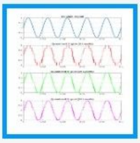
Figures SHOW DATA CURSOR

Figure 1: dsp6_1





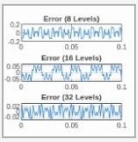


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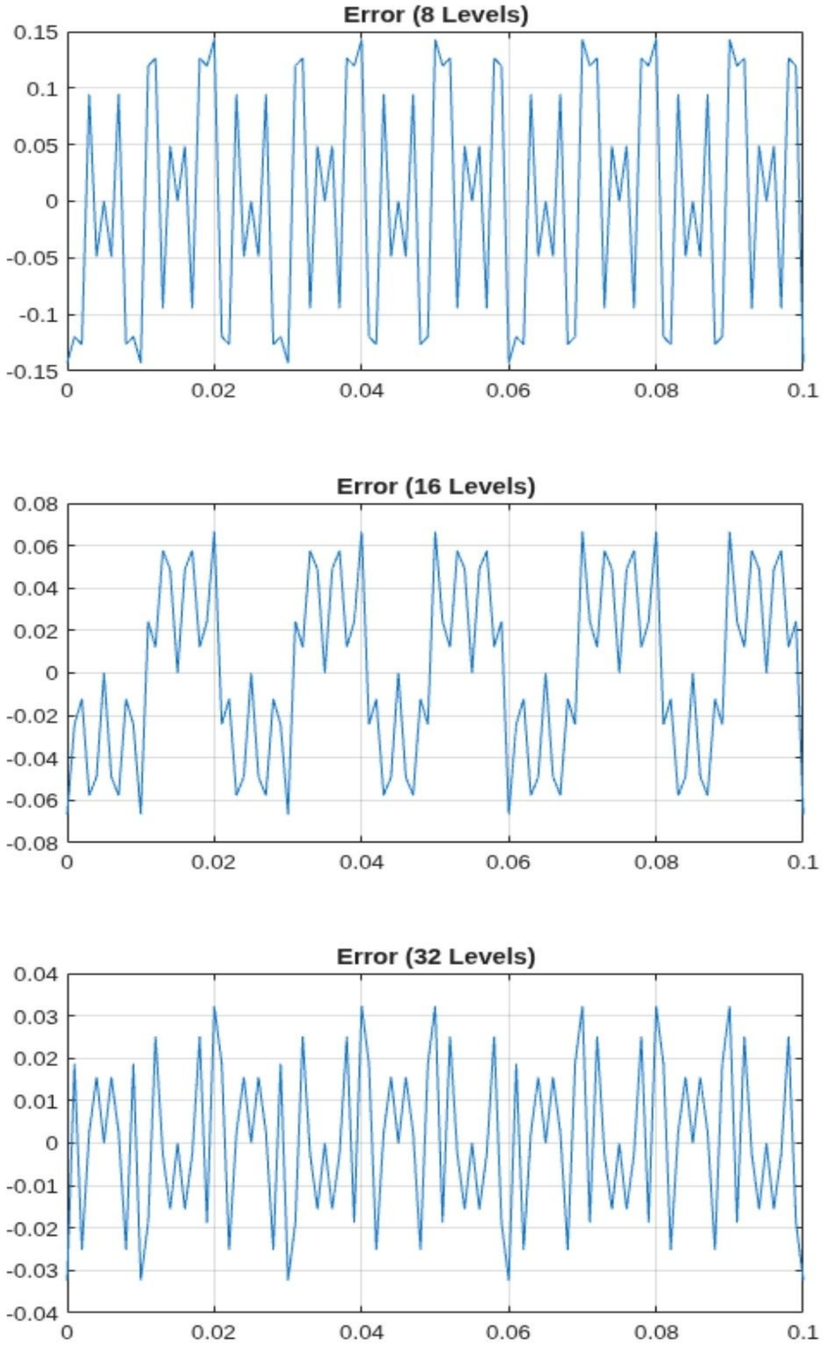
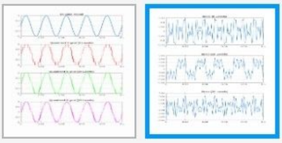
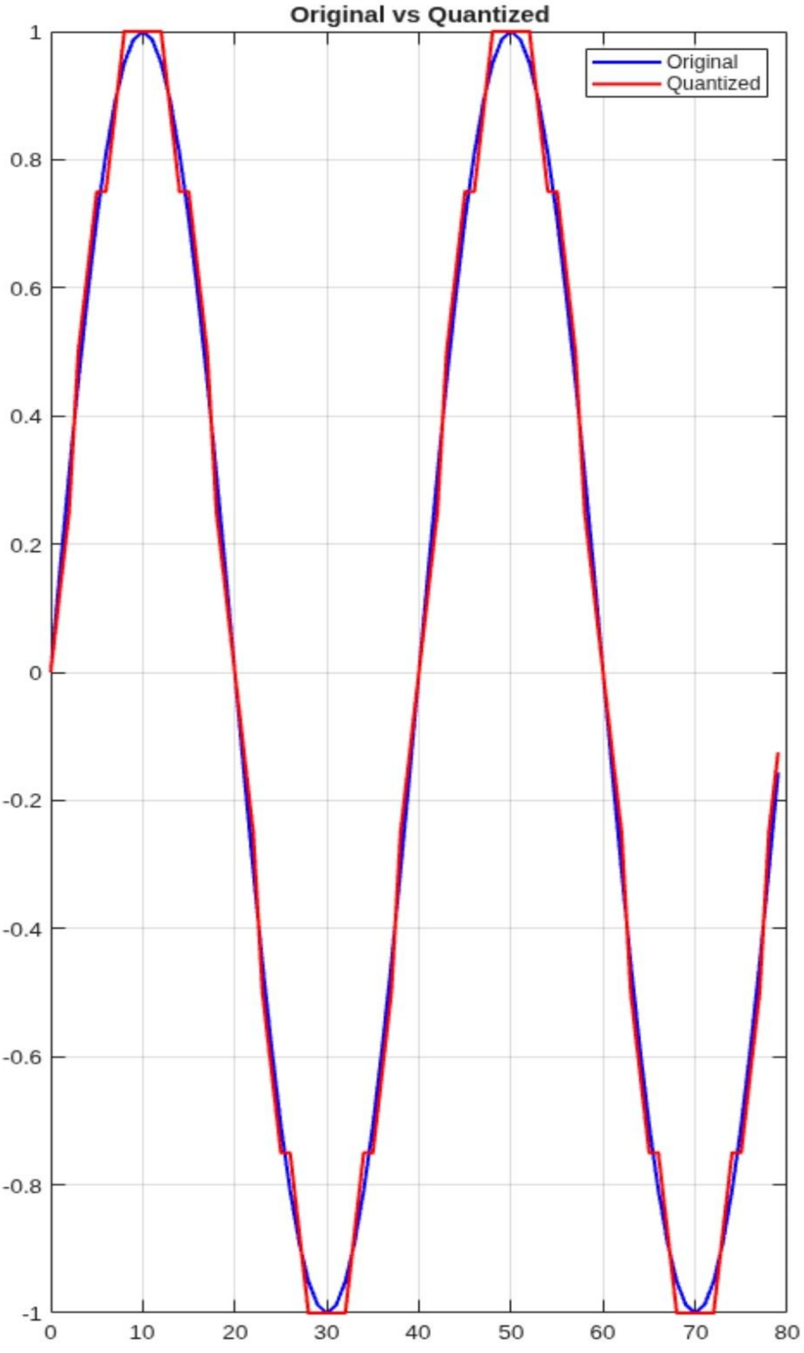
Figure 2: dsp6_1





```
clc;
clear;
close all;

n = 0:79;
x = sin(2*pi*n/40);

q = 8;

xt = floor(x*q)/q;
xr = round(x*q)/q;

et = x - xt;
er = x - xr;

% Fig 1
figure;
plot(n,x,'b',n,xr,'r','LineWidth',1.5);
grid on;
title('Original vs Quantized');
legend('Original','Quantized');

% Fig 2
figure;
plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5
grid on;
title('Truncation vs Rounding');
legend('Original','Truncated','Rounded');

% Fig 3
figure;
stem(n,et,'r');
hold on;
stem(n,er,'g');
grid on;
title('Quantization Error');
legend('Truncation','Rounding');

% Fig 4
figure;
bar([mean(et.^2) mean(er.^2)]);
set(gca,'XTickLabel',{'Truncation','Rounding'}
grid on;
title('MSE Comparison');
ylabel('MSE');
```

Figure 1: dsp6_2



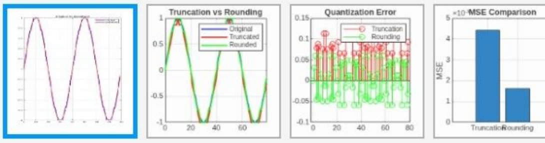


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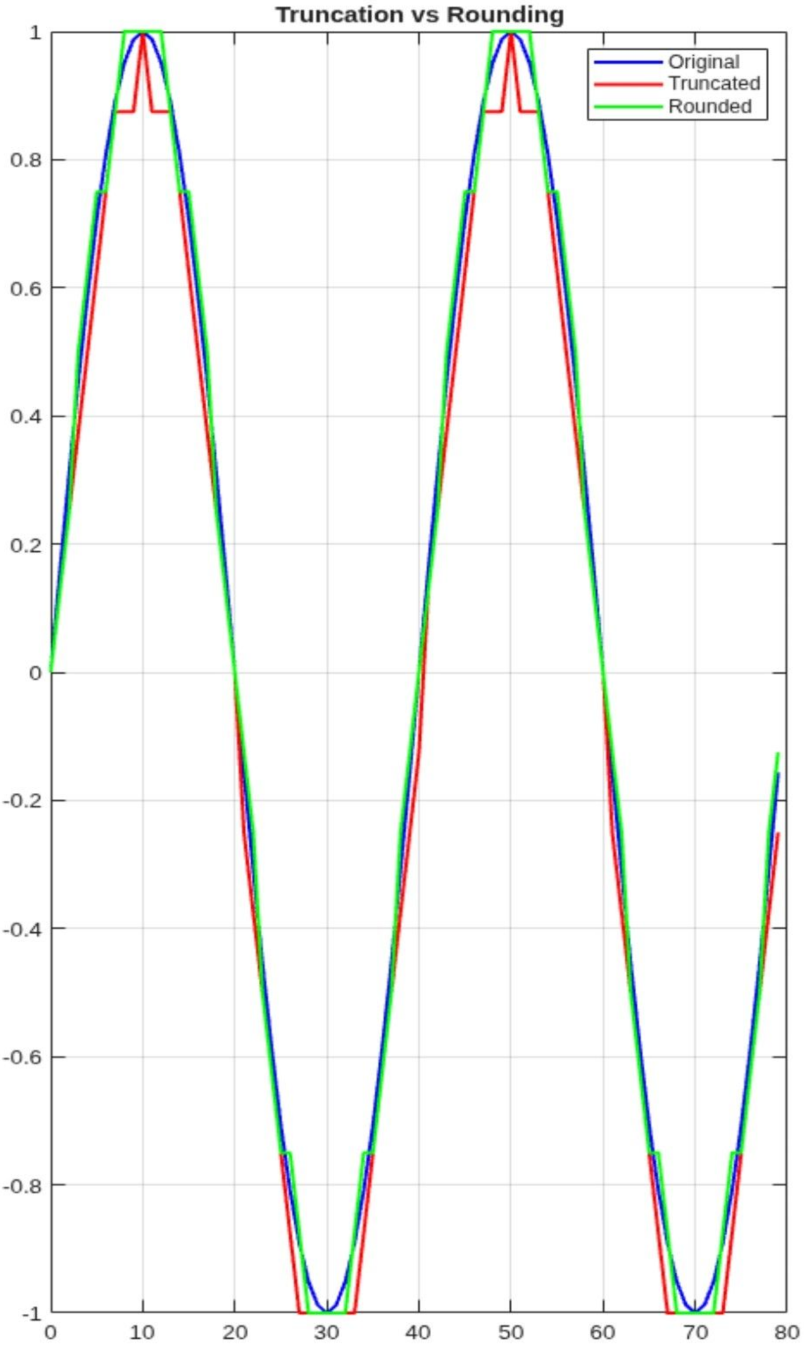
Figure 2: dsp6_2



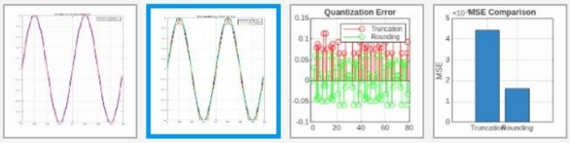


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Figure 3: dsp6_2

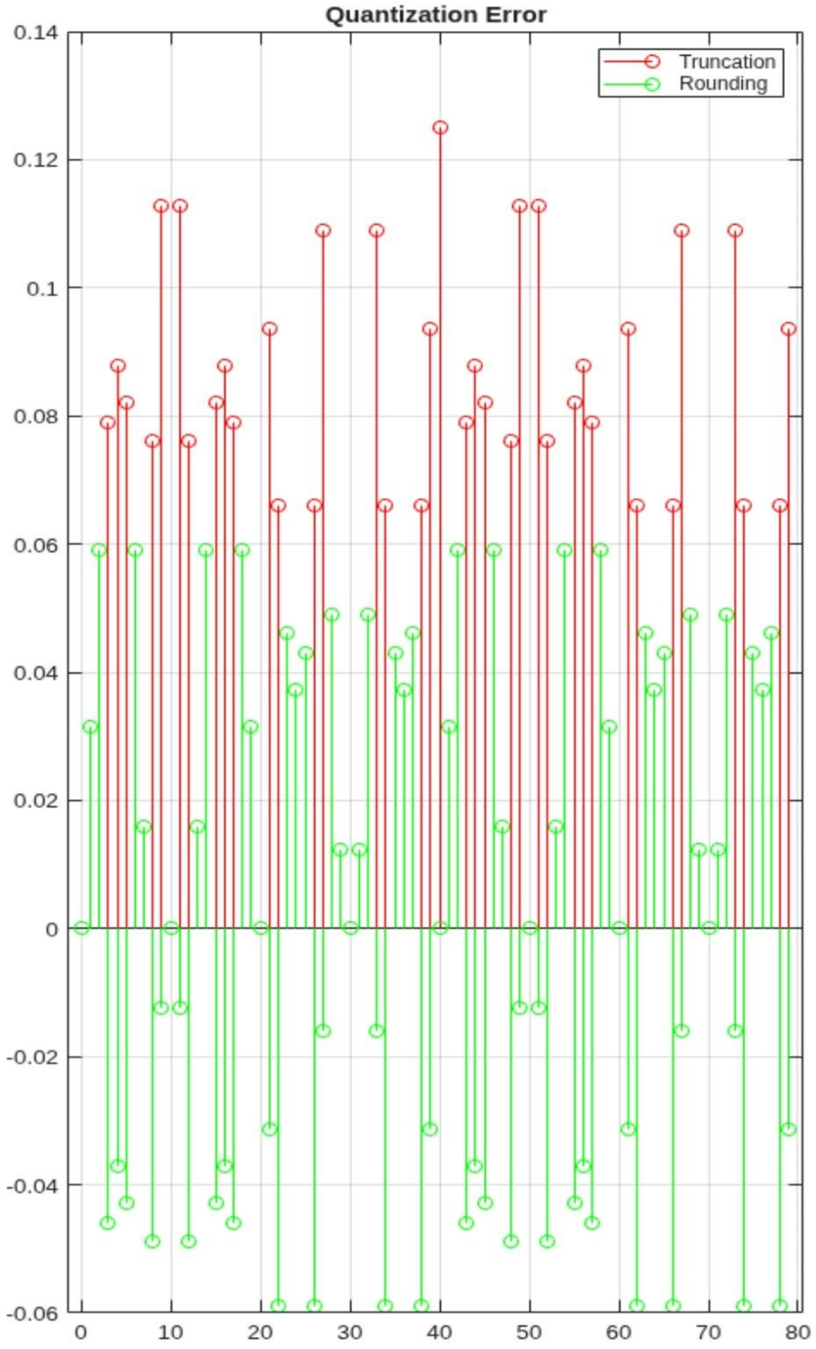

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Figure 4: dsp6_2